



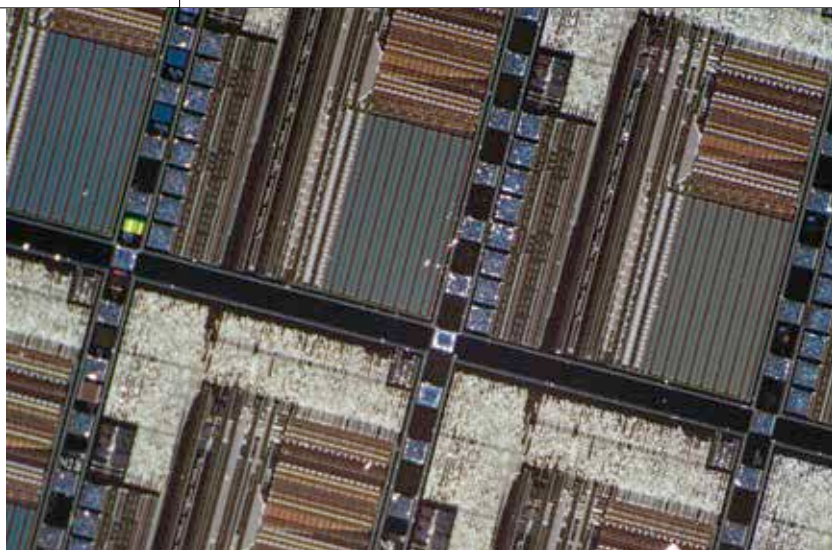
Cutting down on forwards

WhatsApp is taking steps to cut down on circulation of viral message forwards by imposing a limit. <https://digit.in/may20-27>



Ease sale of non-essentials

Amazon and Flipkart are requesting the govt to ease sale of non-essential products in the country. <https://digit.in/may20-28>



THE HUMBLE TRANSISTOR

The history of what may be the most important invention of the modern world

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odern computers are continually revolutionising our lives by performing tasks that were deemed almost

impossible decades ago. A long array of innovations has allowed modern computers, as we know them today, to come to pass. However, there's one foundational invention that almost everything else relies upon. The humble transistor. Let's see how this inconspicuous device came to be.

THE NEED FOR TRANSISTORS

At their very core, all computers are fundamentally machines that perform mathematical operations. The earliest ones were manual counting devices, such as the abacus, while the

ones developed in the future utilised mechanical parts. What made these devices 'computers' was having a way to successfully represent numbers and a system to manipulate the numbers. Electronic computers work the same way, however, instead of physical arrangements, the numbers are depicted in electric voltages. Boolean logic, that has only two possible values – true (1) and false (0), comes to play in electronic computers. In electronic computers, equations are performed via logic gate circuits that produce an output of 1 or 0. For more advanced operations, these circuits are fused



ENIAC used vacuum tubes to control electric voltages

together. Computer programs consist of sets of instructions which detail how to perform these operations. However, this system needs a reliable and accurate method for controlling the electric current.

Early electronic computers such as the ENIAC (Electrical Numerical Integrator And Calculator), developed in 1946, used a device known as the vacuum tube to control the electric voltages. Vacuum tubes used in such computers were bulky and unreliable. They also consumed absurd amounts of power and generated too much heat which, in turn, shortened the lifespan of these tubes. The transistor, on the other hand, was much smaller and used considerably lesser power than the vacuum tubes. Also, since the transistor used lesser power, the amount of heat generated is minimal. Therefore, it doesn't fail quite as often as vacuum tubes. The transistor uses a semiconductor instead of electrodes which were used in vacuum tubes.

A semiconductor such as silicon is treated with different elements such as phosphorus and boron to create an electron-emitting 'N-Type' and an electron-absorbing 'P-Type'. These are arranged as N-P-N (typical NPN transistor), with a terminal placed at each – the emitter, the base and the collector. In such transistors, small variations in the input voltage can be used to rapidly switch between high and low voltages. The transistor has allowed the development of modern-day computers. ENIAC was a massive computer which weighed a whopping 30 tons and was the size of a tennis court! In contrast, ENIAC's functionality can now be surpassed by a single, fingernail-sized microprocessor containing billions of transistors.

THE DRIVE TOWARDS AN ALTERNATIVE SOLUTION

In the 1920s, researchers began investigating the use of semiconductors for amplifying and switching signals in telephonic devices. In the 1930s, Marvin Kelly, an American physicist and engineer, at Bell Labs decided to